Final Exam

● Graded

Student

Kritagya Paudel

Total Points

41.5 / 45 pts

Question 1

Overall Grades **41.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 3.5 pts** Point adjustment

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
KRITAGYA PAUDEL	kp430	

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

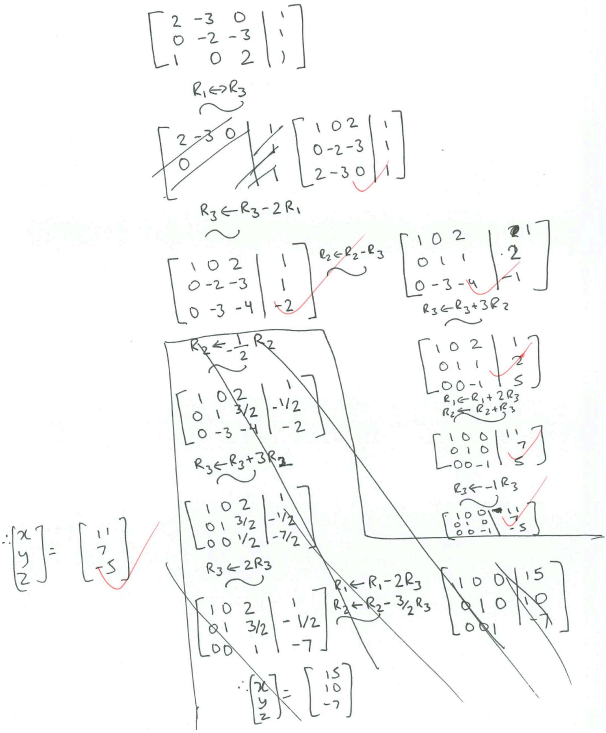
Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: 

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	4.5	5	5	2	5	5	5	5	41.5

Problem 1. [5] Find the general solution of the linear system:

$$2x - 3y = 1$$
$$-2y - 3z = 1$$
$$x + 2z = 1$$



4.5/7

Problem 2. [5] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

$R_3 \leftrightarrow R_2$, $R_4 \leftrightarrow R_1$

$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ -5 & 0 & 5 \\ 2 & 4 & 2 \end{bmatrix}$

$R_3 \leftarrow R_3 + 5R_1$, $R_4 \leftarrow R_4 - 2R_1$

$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 15 & 15 \\ 0 & -2 & -2 \end{bmatrix}$

$R_3 \leftarrow \frac{1}{15} R_3$, $R_4 \leftarrow -\frac{1}{3} R_4$

$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$R_4 \leftarrow R_4 - R_2$, $R_3 \leftarrow R_3 - R_2$

$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$R_1 \leftarrow R_1 - 3R_2$

$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(1) basis of column space of $A = \left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

(2) basis of row space of $A = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ -0.5

(3) basis of null space of $A = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

$x - z = 0$
$x = z$
let $z = s$
$y + z = 0$
$\therefore y = -s$

$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Check

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$T(x) = Ax$, let $A = \begin{bmatrix}a & b & c\\ d & e & f\end{bmatrix}$

$$T\begin{bmatrix}0\\1\\-1\end{bmatrix} = \begin{bmatrix}a & b & c\\ d & e & f\end{bmatrix}\begin{bmatrix}0\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix} \text{ or, } \begin{bmatrix}b-c\\ e-f\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix} - (1)$$

$$T\begin{bmatrix}1\\0\\-1\end{bmatrix} = \begin{bmatrix}a & b & c\\ d & e & f\end{bmatrix}\begin{bmatrix}1\\0\\-1\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix} \text{ or, } \begin{bmatrix}a-c\\ d-f\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix} - (2)$$

$$T\begin{bmatrix}0\\0\\-1\end{bmatrix} = \begin{bmatrix}a & b & c\\ d & e & f\end{bmatrix}\begin{bmatrix}0\\0\\-1\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix} \text{ or } \begin{bmatrix}-c\\ -f\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix} - (3)$$

$\therefore c = -2$, $f = -2$; $a + 2 = 0 \Rightarrow \therefore a = -2$; $d + 2 = 2 \Rightarrow d = 0$; $b + 2 = 1 \Rightarrow \therefore b = -1$; $e + 2 = 1 \Rightarrow \therefore e = -1$

$$\therefore A = \begin{bmatrix}-2 & -1 & -2\\ 0 & -1 & -2\end{bmatrix}$$

Problem 4. [5'] Find the Jordan canonical form of

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}, \quad \det(A-\lambda I) = (1-\lambda)(1-\lambda)(2-\lambda) = (1-\lambda)^2(2-\lambda)$$

So, $\lambda = 1, \quad \lambda = 2$

Now, For $\lambda = 1$,

$$A - I = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \xrightarrow[R_2 \leftarrow -\frac{1}{2}R_3]{R_1 \leftarrow R_1 - 2R_2} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$x + y = 0$

$y = 0$

$R_1 \leftarrow R_1 - R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$V_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

It's algebraic multiplicity = 2 but geometric multiplicity = 1, so $\lambda = 1$'s jordan block = 2×2.

For $\lambda = 2$,

$$A - 2I = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix} \xrightarrow[R_2 \leftarrow -\frac{1}{2}R_3]{R_1 \leftarrow -R_1} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1/2 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_2} \begin{bmatrix} 1 & -2 & 0 \\ 1 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & -2 & 0 \\ 0 & +3 & 1/2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

p.1? 2/5

Problem 5. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$$\begin{array}{c} \\ x \\ y \\ z \end{array}\begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} x^2 & xy & xz/2 \\ xy & y^2 & yz/2 \\ xz/2 & yz/2 & z^2 \end{bmatrix} \end{array}$$

$$f(x,y,z) = \begin{bmatrix} x & y & z \end{bmatrix} A \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \text{ where } A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}, \quad \det(A-\lambda I) = (3-\lambda)\left[(3-\lambda)(4-\lambda) - 1\right] - 0 + (3-\lambda)$$

$$= (3-\lambda)\left[12 - 3\lambda - 4\lambda + \lambda^2 - 1\right] + (3-\lambda)$$

$$= (3-\lambda)\left(\lambda^2 - 7\lambda + 11\right) + (3-\lambda)$$

$$= 3\lambda^2 - 21\lambda + 33 - \lambda^3 + 7\lambda^2 - 11\lambda + 3 + \lambda$$

$$= -\lambda^3 + 10\lambda^2 - 32\lambda + 36$$

$$= (3-\lambda)(\lambda^2 - 7\lambda + 11 - 1)$$

$$= (3-\lambda)(\lambda^2 - 7\lambda + 10)$$

$$= (3-\lambda)(\lambda - 5)(\lambda - 2)$$

$\lambda = 3, \ \lambda = 5, \ \lambda = 2$

$$A - 3I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \overset{R_2 \leftarrow -R_2}{\overset{R_1 \leftrightarrow R_3}{\sim}} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \overset{R_3 \leftarrow R_3 - R_2}{\sim} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x - y + z = 0 \\ z = 0 \\ x = y \end{array}$$

$$\therefore v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$A - 5I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \overset{R_1 \leftrightarrow R_3}{\sim} \begin{bmatrix} 1 & -1 & -1 \\ 0 & -2 & -1 \\ -2 & 0 & 1 \end{bmatrix} \overset{R_3 \leftarrow R_3 + 2R_1}{\sim} \begin{bmatrix} 1 & -1 & -1 \\ 0 & -2 & -1 \\ 0 & -2 & -1 \end{bmatrix} \overset{R_3 \leftarrow R_3 + R_2}{\sim} \begin{bmatrix} 1 & -1 & -1 \\ 0 & -2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\overset{R_1 \leftarrow R_1 + 2R_3}{\sim} \begin{bmatrix} 1 & -1 & 1 \\ 0 & -2 & -1 \\ 0 & 1 & -1 \end{bmatrix} \overset{R_3 \leftrightarrow R_2}{\sim} \begin{bmatrix} 1 & -1 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 0 \end{bmatrix} \overset{R_2 \leftarrow R_2 + 2R_3}{\sim} \begin{bmatrix} 1 & -1 & 1 \\ 0 & -2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\overset{R_1 \leftarrow R_3 + R_1}{\sim} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & -1 \end{bmatrix} \overset{R_3 \leftarrow R_3 - R_2}{\sim} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{bmatrix} \overset{-\frac{1}{2}R_3}{\sim} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix} \overset{R_2 \leftarrow R_2 - R_3}{\sim} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \overset{-\frac{1}{2}R_2}{\sim} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \overset{R_3 \leftarrow R_3 - R_1}{\sim} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \overset{R_3 \leftarrow R_3 + R_2}{\sim} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore v_2 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

5/5

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$R_3 \leftrightarrow R_1$

$$\left[\begin{array}{ccc|c} -2 & 2 & 0 & -8 \\ 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \end{array}\right]$$

$R_2 \leftarrow R_1 + R_2$

$$\left[\begin{array}{ccc|c} -2 & 2 & 0 & -8 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & -1 & -3 \end{array}\right]$$

$R_1 \leftarrow -\frac{1}{2}R_1$

$R_3 \leftarrow R_3 - 2R_2$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & -3 \end{array}\right]$$

$R_3 \leftarrow -\frac{1}{3}R_3$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$R_2 \leftarrow R_2 - R_3$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$\therefore a = 3, b = -1, c = 1$ //

$R_1 \leftarrow R_1 + R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

$v_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ $\|v_1\| = \sqrt{1+1} = \sqrt{2}$

$\hat{v}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$w_2 = v_2 - \frac{v_2 \cdot v_1}{v_1 \cdot v_1} v_1$

$v_2 \cdot v_1 = \begin{pmatrix}0\\1\\1\\0\end{pmatrix} \cdot \begin{pmatrix}3\\0\\2\\-1\end{pmatrix} = 0+0+2+0 = 2$

$v_1 \cdot v_1 = \begin{pmatrix}0\\1\\1\\0\end{pmatrix} \cdot \begin{pmatrix}0\\1\\1\\0\end{pmatrix} = 2$

$w_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{2}\begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

$\|w_2\| = \sqrt{12} = 2\sqrt{3}$

$\therefore \hat{v}_2 = \frac{1}{2\sqrt{3}}\begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

5/5

Problem 8. [5] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

Let A be 2×2 matrix satisfying $A^T = -A$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2\times 2}, \text{ So, } A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix}$$

~~for, $A^T = -A$~~, Since, $a=-a$, $d=-d$, This is only true when $a=d=0$. ∴ The left to right diagonal in 2×2 matrix = 0.

$\det(A) = -cb$.

Let B be 3×3 matrix satisfying $A^T = -A$.

$$B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}_{3\times 3}, \text{ So, } B^T = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} = \begin{bmatrix} -a & -b & -c \\ -d & -e & -f \\ -g & -h & -i \end{bmatrix}$$

Since, $a=-a$, $e=-e$, $i=-i$, this is only true when $a=e=i=0$. So, the ~~determinant, $\det(B)=0$~~. $\text{tr}(B) = 0$, ∴ $\det(B) = 0$.

$a=e=i=0$
$b=-d$
$c=-g$
$f=-h$

$$\det(B) = +a[ei-hf] + d[bi-ch] + g[bf-ce]$$
$$= 0 - b[ch] - c[bf+0]$$
$$= +cbf - cbf$$
$$= 0$$

Shorter Proof:

$\det(A^T) = \det(A)$

Here, $A^T = -A$,

So, ~~$\det(A)$~~ $\det(-A) = \det(A)$

or, $(-1)^n \det(A) = \det(A)$

or, $\det(A) - (-1)^n \det(A) = 0$

∴ when ~~n is even~~ n is odd {1, 3, 5, ...}

or, $\det(A) + \det(A) = 0$

$2\det(A) = 0$

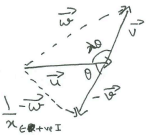
$\therefore \det(A) = 0$

5/5

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.



$(\|\vec{u}+\vec{v}\|)^2 = (\|\vec{u}-\vec{v}\|)^2$

$\|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2$

$2\vec{u}\cdot\vec{v} = -2\vec{u}\cdot\vec{v}$

$4\vec{u}\cdot\vec{v} = 0$

$\therefore \vec{u}\cdot\vec{v} = 0$

$\therefore \vec{u}$ and $\vec{v}$ are perpendicular to each other.

